Open Resources for
Community College Algebra

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Faculty
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Project Lead: Alex Jordan
Contributing Authors: Ann Cary, Alex Jordan, Ross Kouzes, Scot Leavitt, Cara Lee, Carl Yao, Ralf Youtz
WeBWorK Problem Coding: Chris Hughes, Alex Jordan, Carl Yao
Significant Contributors: Kara Colley, Wendy Lightheart, Ian Kit Nicolas
Cover Image: Ralf Youtz

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The technology that makes it possible to create integrated print, HTML, and WeBWorK content is PreTeXt, created by Rob Beezer. David Farmer and several other PreTeXt contributors are credited with the PreTeXt HTML layout being feature-rich and easy to navigate. A grant from OpenOregon funded the original bridge between WeBWorK and PreTeXt.

This book uses WeBWorK to provide most of its exercises, which may be used for online homework. WeBWorK was created in the early 1990s by Mike Gage and Arnie Pizer, and has had decades of contributions from open source developers. In 2013, Chris Hughes, Alex Jordan, and Carl Yao programmed most of the first edition WeBWorK questions with a PCC curriculum development grant.

The Javascript library MathJax, maintained by David Cervone and Volker Sorge, allows math content to render nicely on screen in the HTML version. Additionally, MathJax makes web accessible mathematics possible.

The PDF versions are built using the typesetting software LaTeX, created by Donald Knuth and enhanced by Leslie Lamport.

Some exercises in this textbook use features developed for Runestone Academy, a hosting platform for interactive open-source textbooks. You may create a course on Runestone Academy that uses this textbook, and track student progress and grades there. Brad Miller is the creator and maintainer of Runestone Academy.

Each of these open technologies, along with many that we use but have not listed here, has been enhanced by many additional contributors over many decades. We are indebted to these contributors for their many contributions. By releasing this book with an open license, we honor their dedication to open software and open education.

To All

HTML, PDF, and print. This book is freely available in the following formats: HTML, ePub, PDF-for-screen, PDF-for-print, and Braille. Additionally, a printed and bound copy is available for purchase at low cost. All versions offer essentially the same content and are synchronized such that cross-references match across versions. (The printed and bound versions omit the appendix exercises.) Information about all versions can be found at `pcc.edu/orcca`.

There are some notable differences between the HTML version, PDF-for-screen, and PDF-for-print.

- The HTML version offers interactive elements, easier navigation than print, and its content is web accessible.
- Two PDF versions can be downloaded. One version is intended to be read on a screen with a PDF viewer. This version retains full color, has its text centered on the page, and includes hyperlinking. The other version is intended for printing on two-sided paper and then binding. Most of its color has been converted with care to grayscale. Text is positioned to the left or right of each page in a manner to support two-sided binding. Hyperlinks have been disabled.
- Printed and bound copies are available for purchase online. Up-to-date information about purchasing a copy should be available at `pcc.edu/orcca`. Contact `orcca-group@pcc.edu` if you have trouble finding the latest version online.

Copying Content. The source files for this book are available through `pcc.edu/orcca'`, and openly licensed for use. However, it may be more convenient to copy certain things directly from the HTML version.

The graphs and other images that appear in this book are available in various file formats. While viewing the book with a web browser, right-click on an image to view it in a new tab. The web browser will reveal the path to the image. In almost all cases, you will initially see an `.svg` version of the image. But editing that extension to `.png`, `.eps`, `.pdf`, and `.tex` will provide access to these other formats.

Mathematical content can be copied from the HTML version. To copy math content into *MS Word*, right-click or control-click over the math content, and click to `Show Math As MathML Code`. Click "Copy to Clipboard". In *Word*, use `Paste Special` to paste the content as `Unformatted Text`. To copy math content into LaTeX source, right-click or control-click over the math content, and click to `Show Math As TeX Commands`.

Tables can be copied from the HTML version and pasted into applications like *MS Word*. However, mathematical content within tables will not always paste correctly without a little extra effort as described above.

Accessibility. The eBook version is intended to meet or exceed web accessibility standards. If you encounter an accessibility issue, please report it.

- All graphs and images should have meaningful alt text that communicates what a sighted person would see, without necessarily giving away anything that is intended to be deduced from the image.
- All math content is rendered using MathJax. MathJax has a contextual menu that can be accessed in several ways, depending on what operating system and browser you are using. The most common way is to right-click or control-click on some piece of math content.
- In the MathJax contextual menu, you may set options for triggering a zoom effect on math content, and also by what factor the zoom will be. Also in the MathJax contextual menu, you can activate more accessibility tools, including the ability to display speech strings for math expressions. Setting the speech strings to the "Clearspeak" rules produces speech strings that are intended to match how the math would be spoken aloud.
- A screen reader will generally have success verbalizing the math content from MathJax. With certain screen reader and browser combinations, you may need to set some configuration settings in the MathJax contextual menu.

Tablets and Smartphones. PreTeXt documents like this book are "mobile-friendly". When you view the HTML version, the display adapts to whatever screen size or window size you are using. A math teacher will usually recommend that you do not study from the small screen on a phone, but if it's necessary, the HTML version gives you that option.

WeBWorK for Online Homework. Most exercises are available in a ready-to-use collection of WeBWorK problem sets. Anyone interested in using these problem sets may contact the project leads. The WeBWorK set definition files and supporting files should be available for download from `www.pcc.edu/orcca`.

Odd Answers. The answers to exercises are not printed in the PDF versions for economy. But the HTML version offers answers to odd-numbered exercises in an appendix.

Interactive and Static Examples. Traditionally, a math textbook has examples throughout each section. This textbook uses two types of "example":

Static These are labeled "Example". Static examples may or may not be subdivided into a "statement" followed by a walk-through solution. This is basically what traditional examples from math textbooks do.

Active These are labeled "Checkpoint". In the HTML version, active examples have WeBWorK answer blanks where a reader may try submitting an answer. In the PDF output, active examples are almost indistinguishable from static examples, but there is a WeBWorK icon indicating that a reader could interact more actively using the HTML version. Most of the time, a walk-through solution is provided immediately following the answer blank.
Some readers using the HTML version will skip the opportunity to try an active example and go straight to its solution. That is OK. Some readers will try an active example once and then move on to just read the solution. That is also OK. Some readers will tough it out for a period of time and resist reading the solution until they answer the active example themselves.
For readers of the PDF, the expectation is to read the example and its solution just as they would read a static example.
A reader is *not* required to try submitting an answer to an active example before moving on. A reader *is* expected to read the solution to an active example, even if they succeed on their own at finding an answer.

Reading Questions. Each section has a few "reading questions" immediately before the exercises. These may be treated as regular homework questions, but they are intended to be something more. The intention is that reading questions could be used in certain classroom models as a tool to encourage students to do their assigned reading, and as a tool to measure what basic concepts might have been misunderstood by students following the reading.

Alternative Video Lessons. Most sections open with an alternative video lesson (that is visible in the HTML version and linked using a QR code in PDF). These video playlists are managed through a YouTube account, and it is possible to swap videos out for better ones at any time, provided that does not disrupt courses at PCC. Please contact us if you would like to submit a different video into these video collections.

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Part I

Linear Equations and Lines

Part II

Preparation for STEM

Chapter 1

Solving Quadratic Equations

1.1 Solving Quadratic Equations by Using a Square Root

In this section, we learn how to solve certain types of quadratic equations by using a square root. We will also learn about the Pythagorean Theorem, which is used to find a right triangle's side length when the other two lengths are known.

Video Lessons

1.1.1 Solving Quadratic Equations Using the Square Root Property

When we learned how to solve linear equations, we used inverse operations to isolate the variable. For example, we use *subtraction* to remove an unwanted term that is *added* to one side of a linear equation. When a variable appears in an equation and it is *squared*, we might be able to do something similar and use a *square root* to help find the solution(s). Taking the square root is the inverse of squaring if you happen to know the original number was positive. Even if you don't know that, we can still come to a conclusion about solutions to certain types of equation where the variable is squared.

For example, if $x^2 = 9$, we can think of undoing the square with a square root, and we know $\sqrt{9} = 3$. However, there are actually *two* numbers that we can square to get 9: -3 as well as 3. Both are solutions, and both are in the solution set. This demonstrates something called the Square Root Property.

Fact 1.1.2 Square Root Property. *If k is positive and $x^2 = k$, then*

$$x = -\sqrt{k} \qquad \text{or} \qquad x = \sqrt{k}.$$

*It is common to write $x = \pm\sqrt{k}$ for short, but it is important to remember that this means x could possibly be one of two things: $\sqrt{k}$ or $-\mathrm{sqrt}k$. It does not mean that x is two things at the same time. The positive solution, $\sqrt{k}$, is called the **principal square root** of k.*

Example 1.1.3 Solve for y in $y^2 = 49$.

Explanation.

$$\begin{aligned} y^2 &= 49 \\ y &= \pm\sqrt{49} \\ y &= \pm 7 \end{aligned}$$

$$y = -7 \qquad \text{or} \qquad y = 7$$

To check these solutions, we substitute -7 and 7 in for y in the original equation:

$$\begin{aligned} y^2 &= 49 & y^2 &= 49 \\ (-7)^2 &\stackrel{?}{=} 49 & (7)^2 &\stackrel{?}{=} 49 \\ 49 &\stackrel{\checkmark}{=} 49 & 49 &\stackrel{\checkmark}{=} 49 \end{aligned}$$

Both potential solutions work, and we conclude that the solution set is $\{-7, 7\}$.

Remark 1.1.4 Every solution to a quadratic equation can be checked, as was done in Example 3. In the examples that follow, we won't check solutions. But you should check solutions when solving exercises for your homework or exams.

Checkpoint 1.1.5 Solve for z in $4z^2 - 81 = 0$.

Explanation. Before we can use the square root property we need to isolate the squared quantity.

$$\begin{aligned} 4z^2 - 81 &= 0 \\ 4z^2 &= 81 \\ z^2 &= \frac{81}{4} \end{aligned}$$

Now we can use the square root property

$$\begin{aligned} z &= \pm\sqrt{\frac{81}{4}} \\ z &= \pm\frac{9}{2} \end{aligned}$$

$$z = -\frac{9}{2} \quad \text{or} \quad z = \frac{9}{2}$$

The solution set is $\left\{-\frac{9}{2}, \frac{9}{2}\right\}$.

We can also use the square root property to solve an equation that has a squared *expression* (as opposed to an equation that only has a squared variable).

Example 1.1.6 Solve for p in $50 = 2(p-1)^2$.

Explanation. It's important here to suppress any urge you may have to expand the squared binomial. Perhaps you recently learned how to do that and practiced that skill. Good! But it would not be helpful to do that here. We will begin by isolating the squared expression.

$$\begin{aligned} 50 &= 2(p-1)^2 \\ \frac{50}{2} &= \frac{2(p-1)^2}{2} \\ 25 &= (p-1)^2 \end{aligned}$$

Now that we have the squared expression isolated, we can use the square root property.

$$\begin{aligned} p-1 &= \pm\sqrt{25} \\ p-1 &= \pm 5 \\ p &= \pm 5 + 1 \end{aligned}$$

$$\begin{aligned} p &= -5+1 & \text{or} \quad && p &= 5+1 \\ p &= -4 & \text{or} \quad && p &= 6 \end{aligned}$$

The solution set is $\{-4, 6\}$.

This method of solving quadratic equations is not limited to equations that have rational solutions, or solutions where the radicands are perfect squares. Here are a few examples where the solutions are irrational numbers.

Checkpoint 1.1.7 Solve for q in $(q+2)^2 - 12 = 0$.

Explanation. It's important to suppress any urge to expand the squared binomial.

$$
\begin{aligned}
(q+2)^2 - 12 &= 0 \\
(q+2)^2 &= 12 \\
q+2 &= \pm\sqrt{12} \\
q+2 &= \pm\sqrt{4\cdot 3} \\
q+2 &= \pm 2\sqrt{3} \\
q &= \pm 2\sqrt{3} - 2
\end{aligned}
$$

$$q = -2\sqrt{3} - 2 \quad \text{or} \quad q = 2\sqrt{3} - 2$$

The solution set is $\left\{-2\sqrt{3}-2, 2\sqrt{3}-2\right\}$.

If we solve an equation like the ones we have been solving and end up with a radical in the denominator, we may be expected to rationalize it as covered in `[cross-reference to target(s) "section-rationalizing-the-denominator" missing or not unique]`. We will rationalize a denominator in the next example.

Example 1.1.8 Solve for n in $2n^2 - 3 = 0$.

Explanation.

$$
\begin{aligned}
2n^2 - 3 &= 0 \\
2n^2 &= 3 \\
n^2 &= \frac{3}{2} \\
n &= \pm\sqrt{\frac{3}{2}} \\
n &= \pm\sqrt{\frac{6}{4}} \\
n &= \pm\frac{\sqrt{6}}{2}
\end{aligned}
$$

$$n = -\frac{\sqrt{6}}{2} \qquad \text{or} \qquad n = \frac{\sqrt{6}}{2}$$

The solution set is $\left\{-\frac{\sqrt{6}}{2}, \frac{\sqrt{6}}{2}\right\}$.

Recall that we have no meaning for the square root of a negative number. (Later we will give meaning to that idea, but for now it is meaningless). So if the solving process we are using leads to the square root of a negative number, the original equation has no real solution. Here is an example of an equation with no real solution.

Example 1.1.9 Solve for x in $x^2 + 49 = 0$.

Explanation.

$$\begin{aligned} x^2 + 49 &= 0 \\ x^2 &= -49 \end{aligned}$$

Since $\sqrt{-49}$ is not a real number, we say the equation has no real solution.

1.1.2 The Pythagorean Theorem

Right triangles have an important property called the **Pythagorean Theorem**.

Theorem 1.1.10 The Pythagorean Theorem. *For any right triangle, the lengths of the three sides have the following relationship:*

$$a^2 + b^2 = c^2$$

*where a and b are the lengths of the shorter sides that meet the right angle (referred to as **legs**), and c is the length of the longest side c (which is known as the **hypotenuse**).*

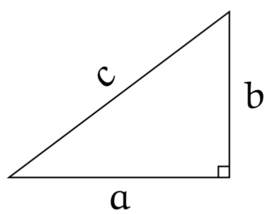


Figure 1.1.11 *In a right triangle, the length of its three sides satisfy the equation* $a^2 + b^2 = c^2$

Example 1.1.12 Find the missing length in this right triangle.

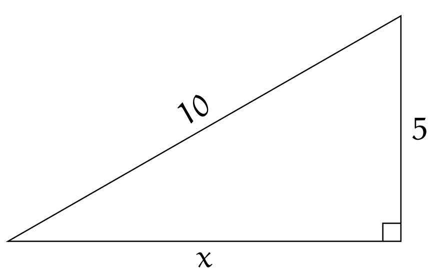


Figure 1.1.13 A Right Triangle

Explanation. We will use the Pythagorean Theorem to solve for x:

$$5^2 + x^2 = 10^2$$

$$
\begin{aligned}
25 + x^2 &= 100 \\
x^2 &= 75 \\
x &= \sqrt{75} && \text{(no need to consider } -\sqrt{75} \text{ in this context)} \\
x &= \sqrt{25 \cdot 3} \\
x &= 5\sqrt{3}
\end{aligned}
$$

The missing length is $x = 5\sqrt{3}$.

Example 1.1.14 Keisha is designing a wooden frame in the shape of a right triangle, as shown in Figure 15. The legs of the triangle are 3 ft and 4 ft. How long should she make the diagonal side? Use the Pythagorean Theorem to find the length of the hypotenuse.

According to Pythagorean Theorem, we have:

$$
\begin{aligned}
c^2 &= a^2 + b^2 \\
c^2 &= 3^2 + 4^2 \\
c^2 &= 25
\end{aligned}
$$

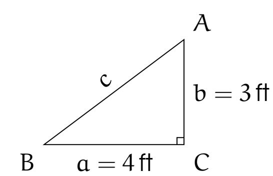


Figure 1.1.15

Now we have a quadratic equation that we need to solve. We need to find the number that has a square of 25. That is what the square root operation does.

$$
\begin{aligned}
c &= \sqrt{25} \\
c &= 5
\end{aligned}
$$

The diagonal side Keisha will cut is 5 ft long.

Note that -5 is also a solution of $c^2 = 25$ because $(-5)^2 = 25$. But a length cannot be a negative number. We need to include both solutions when they are relevant, but also understand when some physical context makes one of the solutions nonsensical.

Example 1.1.16 A 16.5ft ladder is leaning against a wall. The distance from the base of the ladder to the wall is 4.5 feet. How high on the wall does the ladder reach?

The Pythagorean Theorem says:

$$\begin{aligned} a^2 + b^2 &= c^2 \\ 4.5^2 + b^2 &= 16.5^2 \\ 20.25 + b^2 &= 272.25 \end{aligned}$$

Now we need to isolate b^2 in order to solve for b:

$$\begin{aligned} 20.25 + b^2 - 20.25 &= 272.25 - 20.25 \\ b^2 &= 252 \end{aligned}$$

We use the square root property. Because this is a geometric situation we only need to use the principal square root:

$$b = \sqrt{252}$$

Now simplify this radical and then approximate it:

$$\begin{aligned} b &= \sqrt{36 \cdot 7} \\ b &= 6\sqrt{7} \\ b &\approx 15.87 \end{aligned}$$

The ladder reaches about 15.87 feet high on the wall.

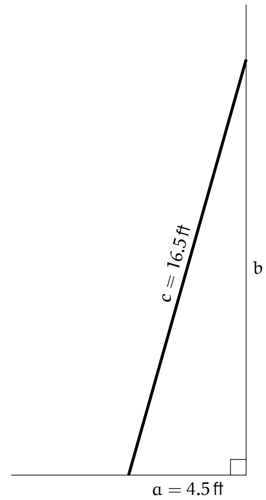


Figure 1.1.17 Leaning Ladder

1.1.3 Reading Questions

1. Typically, how many solutions might there be with a quadratic equation?

2. When you see a $\pm$ sign, as in $x = \pm 2$, is that saying that x is both -2 and 2? Explain.

3. Have you memorized the Pythagorean Theorem? State the formula.

1.1.4 Exercises

Skills Practice

Solving Quadratic Equations with the Square Root Property. Solve the equation.

1. $x^2 = 36$

2. $x^2 = 64$

3. $x^2 = \frac{1}{81}$

4. $x^2 = \frac{1}{100}$

5. $x^2 = 12$

6. $x^2 = 20$

7. $x^2 = 3$

8. $x^2 = 11$

9. $4x^2 = 100$

10. $5x^2 = 45$

11. $x^2 = \frac{9}{100}$

12. $x^2 = \frac{81}{100}$

13. $16x^2 = 49$

14. $64x^2 = 9$

15. $43x^2 - 67 = 0$

16. $19x^2 - 3 = 0$

17. $0 - 2x^2 = -7$

18. $11 - 5y^2 = 8$

19. $17x^2 + 23 = 0$

20. $67x^2 + 31 = 0$

21. $(x+3)^2 = 81$

22. $(x+6)^2 = 25$

23. $(x+8)^2 = 144$

24. $(x+10)^2 = 64$

25. $(4x+2)^2 = 25$

26. $(8x+4)^2 = 121$

27. $1 - 4(y-2)^2 = -3$

28. $4 - 3(y-2)^2 = -8$

29. $(x+7)^2 = 31$

30. $(x-5)^2 = 41$

31. $(t-5)^2 = 20$

32. $(t+10)^2 = 98$

33. $-4 = 41 - (t+4)^2$

34. $-4 = 24 - (x-1)^2$

Isolating a Variable. In the given scenario, isolate the variable that is requested.

35. If you drop an object from height H measured in feet, then after t seconds, its height will be given by $h = H - 16t^2$. Solve this equation for t.

36. If you drop an object from height H measured in meters, then after t seconds, its height will be given by $h = H - 4.9t^2$. Solve this equation for t.

37. The equation for a circle centered at the origin with radius r is $x^2+y^2 = r^2$. Solve this equation for r.

38. The equation for a circle centered at the origin with radius r is $x^2+y^2 = r^2$. Solve this equation for y.

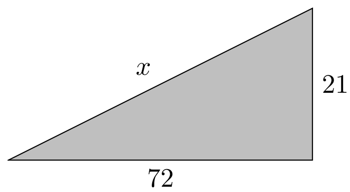
39. The equation for an upward-opening parabola whose vertex is at (h, k) with "curvature" 1 (this is a technical measurement of how curvy a curve is, which you can learn about in calculus) is $y = \frac{1}{2}(x-h)^2 + k$. Solve this equation for x.

40. The equation for an upward-opening parabola whose vertex is at (h, k) with "curvature" 1 (this is a technical measurement of how curvy a curve is, which you can learn about in calculus) is $y = \frac{1}{2}(x-h)^2 + k$. Solve this equation for h.

Applications

Pythagorean Theorem. Find x.

41.



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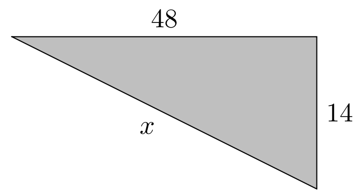

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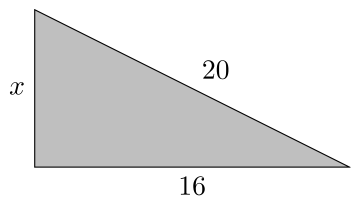

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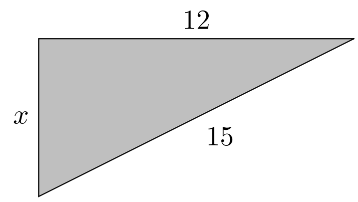

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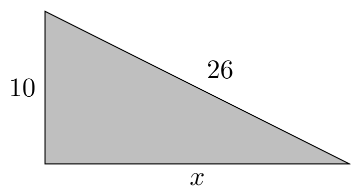

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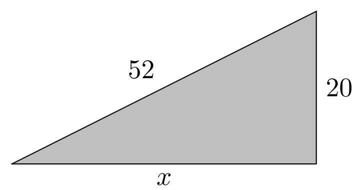

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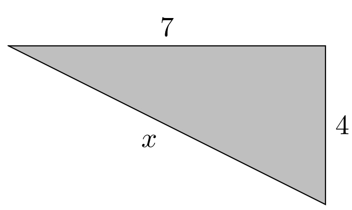

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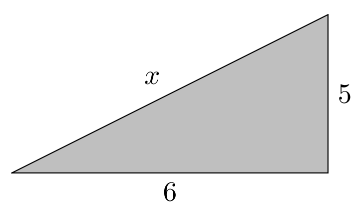

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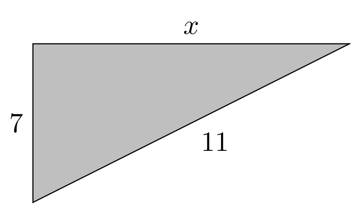

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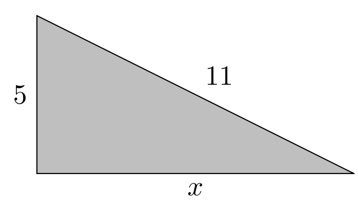


Pythagorean Theorem Applications.

51. Valentina is designing a rectangular garden. The garden's diagonal must be 1 feet, and the ratio between the garden's width and length must be $4:3$. Find the size of the garden's width and length.

52. Aracely is designing a rectangular garden. The garden's diagonal must be 30.6 feet, and the ratio between the garden's width and length must be $15:8$. Find the size of the garden's width and length.

53. Conor is designing a rectangular garden. The garden's width must be 42 feet, and the ratio between the garden's diagonal and length (in the other dimension from its width) must be $13:5$. Find the size of the garden's diagonal and the garden's length.

54. Elliott is designing a rectangular garden. The garden's width must be 4.8 feet, and the ratio between the garden's diagonal and length (in the other dimension from its width) must be $5:3$. Find the size of the garden's diagonal and the garden's length.

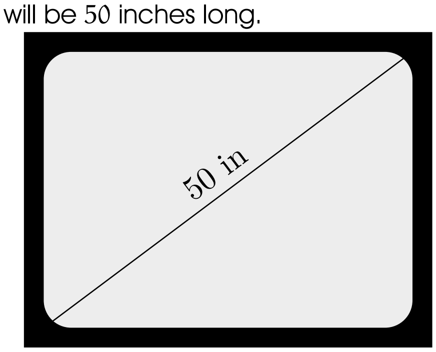
55. A 12-ft ladder is leaning against a wall. The distance from the base of the ladder to the wall is 12 feet. How high on the wall does the ladder reach?

56. A 14-ft ladder is leaning against a wall. The distance from the base of the ladder to the wall is 14 feet. How high on the wall does the ladder reach?

57. To hang some decorations on a home's roof line, you need a ladder that can reach 16 feet. However, the base of the ladder needs to be at least 16 feet removed from direcly below the roofline because of some bushes at the base of the house. Rounding up to the nearest foot, how long of a ladder do you need?

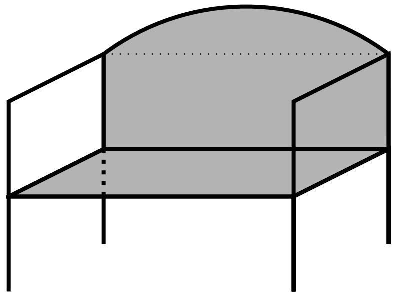
58. To hang some decorations on a home's roof line, you need a ladder that can reach 16 feet. However, the base of the ladder needs to be at least 16 feet removed from direcly below the roofline because of some bushes at the base of the house. Rounding up to the nearest foot, how long of a ladder do you need?

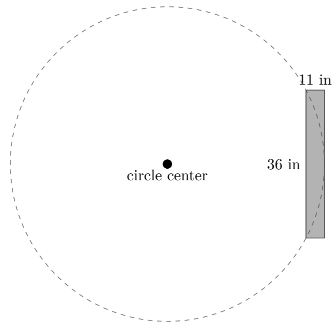
59. Rodney is designing a 50-inch TV, which means that the diagonal of the TV's screen will be 50 inches long.



He needs the screen's width-to-height ratio to be $16:9$. Find the TV screen's width and height.

60. Tyrese wanted to make a bench.





He wanted the top of the bench back to be a perfect portion of a circle, in the shape of an arc, as in the image. (Note that this won't be a half-circle, just a small portion of a circular edge.) He started with a rectangular board 11 inches wide and 36 inches long, and a piece of string, like a compass, to draw a circular arc on the board. How long should the string be so that it can be swung round to draw the arc?

Challenge

61. Imagine that you are in Math Land, where roads are perfectly straight, and Mathlanders can walk along a perfectly straight line between any two points. One day, you bike 2 miles west, 6 miles north, and 4 miles east. Then, your bike gets a flat tire and you have to walk home. How far do you have to walk?

You have to walk ______ miles home.

1.2 The Quadratic Formula

Video Lessons

In Section 1, we solved certain quadratic equations using the square root property. This worked when the equations were in the right format where it was possible to use a square root radical. Now we learn another method to solve quadratic equations in general: the quadratic formula.

1.2.1 Solving Quadratic Equations with the Quadratic Formula

The standard form for a quadratic equation is

$$ax^2 + bx + c = 0$$

where a is some nonzero number.

When $b = 0$ and the equation's form is $ax^2 + c = 0$, then we can simply use the square root property to solve it as we did with equations like this in Section 1. But can we solve equations where $b \neq 0$? A general method for doing this is to use the **quadratic formula**.

Fact 1.2.2 The Quadratic Formula. *For any quadratic equation* $ax^2+bx+c = 0$ *with* $a \neq 0$*, the solutions are given by*

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

As we have seen from solving quadratic equations in Section 1, there can be two solutions to a quadratic equation. Both solutions are represented in the quadratic formula because of the $\pm$ symbol. We could write the two solutions separately:

$$x = \frac{-b - \sqrt{b^2 - 4ac}}{2a} \qquad \text{or} \qquad x = \frac{-b + \sqrt{b^2 - 4ac}}{2a}$$

This method for solving quadratic equations will work to solve *every* quadratic equation. It is most helpful when $b \neq 0$.

Example 1.2.3 Linh is in a physics class that launches a tennis ball from a rooftop that is 90.2 feet above the ground. They fire it directly upward at a speed of 14.4 feet per second and measure the time it takes for the ball to hit the ground below. We can model the height of the tennis ball, h, in feet, with the quadratic equation $h = -16t^2 + 14.4t + 90.2$, where x represents the time in seconds after the launch. According to the model, when should the ball hit the ground?

The ground has height 0 feet, so we should substitute 0 in for h. This gives us the quadratic equation:

$$0 = -16t^2 + 14.4t + 90.2$$

We cannot solve this equation with the square root property, so we will use the quadratic formula. First we will identify that $a = -16$, $b = 14.4$, and $c = 90.2$, and substitute these into the quadratic formula:

$$\begin{aligned} t &= \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} \\ t &= \frac{-(14.4) \pm \sqrt{(14.4)^2 - 4(-16)(90.2)}}{2(-16)} \\ t &= \frac{-14.4 \pm \sqrt{5980.16}}{-32} \end{aligned}$$

These are the exact solutions but because we have a context we want to approximate the solutions with decimals.

$$t \approx -1.966 \text{ or } t \approx 2.866$$

We don't use the negative solution because a negative time does not make sense in this context. The ball will hit the ground approximately 2.866 seconds after it is launched.

The quadratic formula can be used to solve any quadratic equation, but it requires that you remembering the formula correctly and that you correctly identify a, b, and c. Also, that you don't make any arithmetic mistakes when you simplify. We recommend that you always check if you could use the square root property before using the quadratic formula.

Example 1.2.4 Solve for x in $2x^2 - 9x + 5 = 0$.

Explanation. First, we check and see that we cannot use the square root property (because $b \neq 0$) so we will use the quadratic formula. We identify that $a = 2$, $b = -9$ and $c = 5$. Substituting these into the quadratic formula:

$$\begin{aligned} x &= \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} \\ x &= \frac{-(-9) \pm \sqrt{(-9)^2 - 4(2)(5)}}{2(2)} \\ x &= \frac{9 \pm \sqrt{81 - 40}}{4} \\ x &= \frac{9 \pm \sqrt{41}}{4} \end{aligned}$$

This is fully simplified because we cannot simplify $\sqrt{41}$ or reduce the fraction. The solution set is $\left\{ \frac{9-\sqrt{41}}{4}, \frac{9+\sqrt{41}}{4} \right\}$. In this exercise, there is no reason to approximate these with decimals.

When a quadratic equation is not in standard form we must convert it before we can clearly identify the values of a, b, and c.

Example 1.2.5 Solve for x in $x^2 = -10x - 3$.

Explanation. First, we convert the equation into standard form by adding $10x$ and 3 to each side of the equation:

$$x^2 + 10x + 3 = 0$$

Next, we check that we cannot use the square root property so we will use the quadratic formula. We identify that $a = 1$, $b = 10$ and $c = 3$. Substituting them into the quadratic formula:

$$\begin{aligned}
x &= \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} \\
x &= \frac{-10 \pm \sqrt{(10)^2 - 4(1)(3)}}{2(1)} \\
x &= \frac{-10 \pm \sqrt{100 - 12}}{2} \\
x &= \frac{-10 \pm \sqrt{88}}{2}
\end{aligned}$$

We notice that the radical can be simplified:

$$\begin{aligned}
x &= \frac{-10 \pm 2\sqrt{22}}{2} \\
x &= \frac{-10}{2} \pm \frac{2\sqrt{22}}{2} \\
x &= -5 \pm \sqrt{22}
\end{aligned}$$

The solution set is $\{-5 - \sqrt{22}, -5 + \sqrt{22}\}$.

Remark 1.2.6 The irrational solutions to quadratic equations can be checked, but doing this sometimes takes a lot of simplification and is not shown throughout this section. As an example of how much effort goes into a direct check, we will check the solution of $-5 + \sqrt{22}$ from Example 5. We need to replace x with $-5 + \sqrt{22}$ and check that the two sides of the equation are equal. This check is shown here:

$$\begin{aligned}
x^2 &= -10x - 3 \\
(-5 + \sqrt{22})^2 &\stackrel{?}{=} -10(-5 + \sqrt{22}) - 3 \\
(-5)^2 + 2(-5)(\sqrt{22}) + (\sqrt{22})^2 &\stackrel{?}{=} -10(-5 + \sqrt{22}) - 3 \\
25 - 10\sqrt{22} + 22 &\stackrel{?}{=} 50 - 10\sqrt{22} - 3 \\
47 - 10\sqrt{22} &\stackrel{\checkmark}{=} 47 - 10\sqrt{22}
\end{aligned}$$

It worked, and it took a lot of effort. As an alternative, we could use a calculator to find $-5 + \sqrt{22} \approx -0.3096$ and see if this decimal approximation appears to be close to working.

$$\begin{aligned} x^2 &= -10x - 3 \\ (-0.3096)^2 &\stackrel{?}{=} -10(-0.3096) - 3 \\ 0.09585216 &\stackrel{?}{=} 0.096 \end{aligned}$$

These are close. So with the help of a calculator, checking like this can give reasonable confidence that the solution we found is valid.

When the radicand from the quadratic formula, $b^2 - 4ac$, which is called the **discriminant**, is a negative number, the quadratic equation has no real solution.

Example 1.2.7 Solve for y in $y^2 - 4y + 8 = 0$.

Explanation. Identify that $a = 1$, $b = -4$ and $c = 8$. Substitute them into the quadratic formula:

$$\begin{aligned} y &= \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} \\ &= \frac{-(-4) \pm \sqrt{(-4)^2 - 4(1)(8)}}{2(1)} \\ &= \frac{4 \pm \sqrt{16 - 32}}{2} \\ &= \frac{4 \pm \sqrt{-16}}{2} \end{aligned}$$

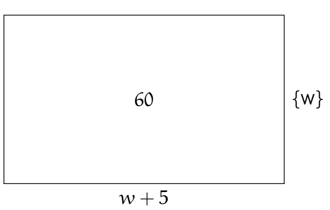
The discriminant worked out to be -16, which is negative. The square root of a negative number is not a real number, so we will conclude that this equation has no real solutions.

1.2.2 Applications

Certain "word problems" lead to a quadratic equation where the quadratic formula may be useful.

Example 1.2.8 A rectangle is 5 inches longer than it is wide. The total area of the rectangle is 60 square inches. How wide is the floor? (This is asking for the shorter dimension.)

Explanation. The area of a rectangle is given by multipying its two dimensions. If we let w stand for the width of the rectangle (the shorter dimension), then $w+5$ is the length, and the total area is $w(w + 5)$. A diagram can help make more clear what needs to be done.



So we must solve the equation $w(w+5)=60$. After multiplying and moving to standard form, we have $w^2+5w-60=0$. Using the quadratic formula:

$$\begin{aligned} w &= \frac{-b \pm \sqrt{b^2-4ac}}{2a} \\ &= \frac{-5 \pm \sqrt{5^2-4(1)(-60)}}{2(1)} \\ &= \frac{-5 \pm \sqrt{265}}{2} \\ w \approx -10.64 &\quad \text{or} \quad w \approx 5.64 \end{aligned}$$

Only the solution $w \approx 5.64$ makes sense as the width of a rectangle. So the rectangle is about 5.64 inches wide.

Checkpoint 1.2.9 Two numbers have sum 35, and their product is -30. Find these two numbers.

Explanation. Let x be one of these numbers. The statement that "the two numbers' sum is 35" means that the other number must be $35-x$. Since the two numbers have product -30. we have the equation

$$\begin{aligned} x(35-x) &= -30 \\ 35x - x^2 &= -30 \end{aligned}$$

This is a quadratic equation which can be rewritten into standard form by adding 30 to each side:

$$-x^2+35x+30=0$$

Then, to solve for x, we apply the quadratic formula:

$$\begin{aligned} x &= \frac{-35 \pm \sqrt{35^2-4(-1)(30)}}{2(-1)} \\ &= \frac{-35 \pm \sqrt{1225+120}}{-2} \\ &= \frac{-35 \pm \sqrt{1345}}{-2} \\ &= \frac{35 \pm \sqrt{1345}}{2} \end{aligned}$$

So the two possible solutions for x are:

$$x = \frac{35-\sqrt{1345}}{2} \quad \text{or} \quad x = \frac{35+\sqrt{1345}}{2}$$

Actually these are the two numbers that sum to 35 and multiply to -30.

Example 1.2.10 Amita has a food stand where she sells momo (Nepali dumplings). If she charges p dollars for each momo, she estimates that she'd sell $80{,}000 - 20{,}000p$ over the course of a year. (This reflects how raising the price can lead to fewer people making purchases.) What is the largest price that Amita could set to end up with a revenue of \$70,000 for the year? (Note that revenue is not the same as profit, and we will not be accounting for Amita's expenses.)

Explanation. If Amita sets the price at p dollars per momo, her total revenue will be (number sold) · price, which is $(80000 - 20000p)p$, which simplifies to $80000p - 20000p^2$. We have been asked to set that equal to 70000.

$$\begin{aligned} 80000p - 20000p^2 &= 70000 \\ -20000p^2 + 80000p - 70000 &= 0 \end{aligned}$$

So many zeros! We have the option to divide by 10000.

$$-2p^2 + 8p - 7 = 0$$

$$\begin{aligned} p &= \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} \\ &= \frac{-8 \pm \sqrt{8^2 - 4(-2)(-7)}}{2(-2)} \\ &= \frac{-8 \pm \sqrt{8}}{-4} \\ p &\approx 2.71 \quad \text{or} \quad x \approx 1.29 \end{aligned}$$

The possible solutions are about 2.71 and 1.29. They are both valid solutions. Amita can reach the target revenue of \$70,000 by setting the price per momo to either \$1.29 or \$2.71.

1.2.3 Radical Equations

Sometimes a radical equation gives rise to a quadratic equation, and the quadratic formula can then be useful.

Example 1.2.11 Solve for z in $\sqrt{z} + 2 = z$.

Explanation. We will isolate the radical first, and then square both sides.

$$\begin{aligned} \sqrt{z} + 2 &= z \\ \sqrt{z} &= z - 2 \\ \left(\sqrt{z}\right)^{\boxed{2}} &= (z-2)^{\boxed{2}} \\ z &= z^2 - 4z + 4 \end{aligned}$$

$$\begin{aligned}
0 &= z^2 - 5z + 4 \\
z &= \frac{5 \pm \sqrt{(-5)^2 - 4(1)(4)}}{2} \\
&= \frac{5 \pm \sqrt{25 - 16}}{2} \\
&= \frac{5 \pm \sqrt{9}}{2} \\
&= \frac{5 \pm 3}{2}
\end{aligned}$$

$$\begin{aligned}
z &= \frac{5-3}{2} & &\text{or} & z &= \frac{5+3}{2} \\
z &= 1 & &\text{or} & z &= 4
\end{aligned}$$

There was a step in this process where we squared both sides of an equation. Squaring both sides of an equation can cause false equations to become true. For example $-2 = 2$ is false, but if you square both sides then $4 = 4$ is true. This means that when an equation solving process involves squaring both sides, and we end up with "solutions" like the two z-values above, they are only *potential* solutions. Logically we've whittled down the list of possible solutions to just these two numbers. But we need to actually check whether or not they actually work.

$$\begin{aligned}
\sqrt{1} + 2 &\stackrel{?}{=} 1 & \qquad \sqrt{4} + 2 &\stackrel{?}{=} 4 \\
1 + 2 &\stackrel{\text{no}}{=} 1 & \qquad 2 + 2 &\stackrel{\checkmark}{=} 4
\end{aligned}$$

It turned out that 1 is an **extraneous solution**, but 4 is a valid solution. So the equation has one solution, 4, and the solution set is $\{4\}$.

Example 1.2.12 Solve the equation $\sqrt{2n-6} = 1 + \sqrt{n-2}$ for n.

Explanation. We cannot isolate two radicals at the same time. One of the radicals is already isolated, so we will simply square both sides and then try to clean up what remains.

$$\begin{aligned}
\sqrt{2n-6} &= 1 + \sqrt{n-2} \\
\left(\sqrt{2n-6}\right)^{\boxed{2}} &= \left(1 + \sqrt{n-2}\right)^{\boxed{2}} \\
2n - 6 &= 1^2 + 2\sqrt{n-2} + \left(\sqrt{n-2}\right)^2 \\
2n - 6 &= 1 + 2\sqrt{n-2} + n - 2 \\
2n - 6 &= 2\sqrt{n-2} + n - 1 \\
n - 5 &= 2\sqrt{n-2}
\end{aligned}$$

Note here that we can leave the factor of 2 next to the radical. We will square the 2 also.

$$\begin{aligned}
(n-5)^{2} &= \left(2\sqrt{n-2}\right)^{2} \\
n^2 - 10n + 25 &= 4(n-2) \\
n^2 - 10n + 25 &= 4n - 8 \\
n^2 - 14n + 33 &= 0
\end{aligned}$$

$$\begin{aligned}
n &= \frac{14 \pm \sqrt{14^2 - 4(1)(33)}}{2} \\
&= \frac{14 \pm \sqrt{196 - 132}}{2} \\
&= \frac{14 \pm \sqrt{64}}{2} \\
&= \frac{14 \pm 8}{2}
\end{aligned}$$

$$\begin{aligned}
n &= \frac{14-8}{2} & \text{or} && n &= \frac{14+8}{2} \\
n &= 3 & \text{or} && n &= 11
\end{aligned}$$

So our two potential solutions are 3 and 11. We should now verify that they truly are solutions.

$$\begin{aligned}
\sqrt{2(3)-6} &\stackrel{?}{=} 1 + \sqrt{3-2} & \qquad \sqrt{2(11)-6} &\stackrel{?}{=} 1 + \sqrt{11-2} \\
\sqrt{6-6} &\stackrel{?}{=} 1 + \sqrt{1} & \sqrt{22-6} &\stackrel{?}{=} 1 + \sqrt{9} \\
\sqrt{0} &\stackrel{?}{=} 1 + 1 & \sqrt{16} &\stackrel{?}{=} 1 + 3 \\
0 &\stackrel{\text{no}}{=} 2 & 4 &\stackrel{\checkmark}{=} 4
\end{aligned}$$

So, 11 is the only solution. The solution set is $\{11\}$.

1.2.4 Reading Questions

1. What is the formula for the discriminant?

2. Are there any kinds of quadratic equations where the quadratic formula is not the best tool to use?

3. Given a quadratic equation, will the quadratic formula always lead you to two solutions?

1.2.5 Exercises

Review and Warmup

Exercise Group. Write each fraction in simplified form.

1. **(a)** $\frac{3}{36}$ **(b)** $\frac{-4}{-18}$ **(c)** $\frac{72}{-8}$ **(d)** $\frac{-10}{2}$

2. **(a)** $\frac{33}{36}$ **(b)** $\frac{36}{-12}$ **(c)** $\frac{-40}{10}$ **(d)** $\frac{-10}{-16}$

3. **(a)** $\frac{-15}{12}$ **(b)** $\frac{-16}{-28}$ **(c)** $\frac{24}{14}$ **(d)** $\frac{20}{28}$

4. **(a)** $\frac{16}{24}$ **(b)** $\frac{-3}{-30}$ **(c)** $\frac{52}{18}$ **(d)** $\frac{-57}{24}$

Exercise Group. Evaluate the given expression for the given values.

5. $b^2 - 4ac$ for $a = 4$, $b = 8$, and $c = 3$

6. $b^2 - 4ac$ for $a = 5$, $b = 4$, and $c = 2$

7. $b^2 - 4ac$ for $a = 5$, $b = -6$, and $c = 4$

8. $b^2 - 4ac$ for $a = -5$, $b = -4$, and $c = 2$

Exercise Group. Simplify the radical.

9. $\sqrt{27}$ **10.** $\sqrt{96}$ **11.** $\sqrt{63}$ **12.** $\sqrt{40}$

Skills Practice

Exercise Group. Use the quadratic formula to solve the equation.

13. $x^2 + 14x + 48 = 0$

14. $x^2 + 2x - 35 = 0$

15. $3x^2 - 28x + 49 = 0$

16. $8x^2 + 23x - 36 = 0$

17. $45x^2 + 52x - 32 = 0$

18. $56x^2 - 39x + 4 = 0$

19. $x^2 + 2x + 3 = 0$

20. $x^2 + 2x + 4 = 0$

21. $x^2 + \frac{71}{8}x + 7 = 0$

22. $x^2 - \frac{37}{7}x + \frac{10}{7} = 0$

23. $x^2 - 10x + 25 = 0$

24. $x^2 - 14x + 49 = 0$

25. $x^2 + x + 7 = 0$

26. $x^2 - 2x + 4 = 0$

27. $x^2 - \frac{5}{3}x - \frac{14}{9} = 0$

28. $x^2 + \frac{64}{63}x + \frac{16}{63} = 0$

29. $x^2 + 6x + 4 = 0$

30. $x^2 + 9x + 5 = 0$

31. $x^2 - \frac{2}{3}x + \frac{1}{9} = 0$

32. $x^2 - \frac{5}{2}x + \frac{25}{16} = 0$

33. $x^2 - 9x + 7 = 0$

34. $x^2 + 8x + 9 = 0$

35. $2x^2 - 8x - 8 = 0$

36. $-6x^2 + 2x + 9 = 0$

37. $9x^2 + 12x + 4 = 0$

38. $9x^2 + 12x + 4 = 0$

39. $x^2+\frac{3}{2}x-\frac{1}{2}=0$

40. $x^2+\frac{1}{2}x-\frac{2}{3}=0$

41. $\frac{1}{2}x^2-\frac{1}{2}x-\frac{1}{2}=0$

42. $\frac{1}{3}x^2+x+\frac{1}{2}=0$

43. $x^2-17.8x+79.21=0$

44. $x^2+17.6x+77.44=0$

45. $-2.6x^2-28.6x-78.65=0$

46. $6.8x^2+57.12x+119.952=0$

47. $x^2+11.6x+19.2=0$

48. $x^2-5.8x+7.2=0$

49. $x^2-3.1x+1.7=0$

50. $x^2-6.8x+3.1=0$

51. $6.3x^2+15.1x+7.1=0$

52. $8.6x^2-4.9x-4.4=0$

53. $x^2-10x=-9$

54. $x^2-4x=12$

55. $x^2=-x+12$

56. $x^2=-5x+36$

57. $5x^2-36x=-7$

58. $3x^2+12=20x$

59. $2x^2=-11x+6$

60. $6x^2=-31x-28$

61. $16x^2+74x=-63$

62. $30x^2+41x=-7$

63. $15x^2=17x+42$

64. $27x^2=-48x-16$

65. $x^2+4=-3x$

66. $x^2-2x=-5$

67. $x^2-4=-\frac{15}{2}x$

68. $x^2+\frac{36}{7}=+\frac{67}{7}x$

69. $x^2-\frac{13}{15}x=+\frac{2}{5}$

70. $x^2+\frac{3}{10}x=+\frac{27}{10}$

71. $x^2+4x=8$

72. $x^2-6=3x$

73. $4x^2+1=x$

74. $5x^2+6=5x$

75. $x^2=+\frac{59}{9}x+\frac{28}{9}$

76. $x^2=-\frac{34}{9}x+\frac{8}{9}$

77. $x^2=+\frac{5}{3}x-\frac{2}{3}$

78. $x^2=-2x+\frac{21}{4}$

79. $-3x^2+9x=4$

80. $-8x^2-9=-19x$

81. $x^2=5x+2$

82. $x^2=-7x+4$

83. $-3x^2=5x-3$

84. $2x^2=-3x+6$

Radical Equations That Give Rise to Quadratic Equations. Solve the equation.

85. $5\sqrt{x}=x+6$

86. $6\sqrt{x}=x+8$

87. $2\sqrt{x}=x-8$

88. $\sqrt{x}=12-x$

89. $3\sqrt{x}=2x+4$

90. $3\sqrt{x}=x+5$

91. $4\sqrt{x}=-x-3$

92. $5\sqrt{x}=-x-4$

93. $5\sqrt{3x+4}=3x+8$

94. $\sqrt{5x-6}=x$

95. $6\sqrt{5x+5}-5x=13$

96. $6\sqrt{5x+7}-15=5x$

97. $\sqrt{3x+12}=-3x$

98. $\sqrt{2x+2}+2x=0$

Applications

99. While standing on a rooftop that is 5 m high off the ground, you throw a tennis ball off the roof in an arc. Because of the speed that you throw the ball, its height from the ground, in meters, t seconds after the throw is $-4.9t^2 + 26t + 7$. How long will it be until the ball hits the ground?

100. While standing on a tree branch that is 23 ft high off the ground, a squirrel kicks a nut in an arc. The nut falls cleanly to the ground without hitting anything on the way down. While the nut is in the air, its height from the ground, in feet, t seconds after the kick is $-16t^2 + 3.1t + 23$. How long will it be until the nut hits the ground?

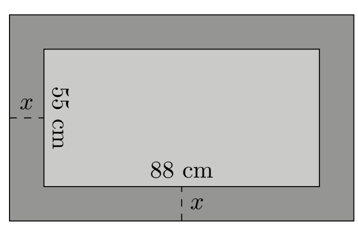
101. An astronaut on the moon makes a running jump off a cliff onto a broad flat plain. The clifftop is 4 m higher than the plain. Because of the slight upward jump, the astronaut's height from the plain, in meters, t seconds after the jump is $-0.81t^2 + 1.2t + 6$. How long will it be until the astronaut lands in the plain?

102. An robot on Mars makes a running jump off a cliff onto a broad flat plain. The clifftop is 15 m higher than the plain. Because of the slight upward jump, the robot's height from the plain, in meters, t seconds after the jump is $-1.85t^2 + 3.9t + 17$. How long will it be until the robot lands in the plain?

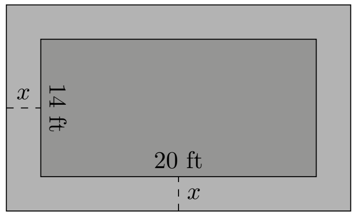
103. One rectangular wall in your classroom is 13.5 ft longer than it is tall. The total area of the wall is 252 ft^2. How tall is the wall?

104. The rectangular floor of your classroom is 9 m longer than it is wide. The total floor area is 220 m^2. How wide is the floor? (This is asking for the shorter dimension.)

105. A metals factory will produce steel picture frames in the form of a rectagle with a smaller rectangle cut out, as in the diagram. The widths of the four edges is to be consistent, and leave a 88 cm by 55 cm visible area for photos and other images. There is a standard thickness for the metal from which these frames will be cut, and the manufacturer has a target weight in mind for the final product so that shipping costs will be kept low. This leads to the decision that the surface area of the frame needs to be 1661 cm^2. How wide should the four edges be? (What is the value of x in the diagram?)



106. A vegetable garden will be surrounded by a frame of grass sod. The garden itself will be 20 ft by 14 ft. The widths of the four edges of the frame will be equal to each other. There is only 177 ft^2 of grass sod available, and it will all be used for this frame. How wide should the four edges be? (What is the value of x in the diagram?)



107. You are ready to self-publish a book online. If the book were free, you estimate 50000 would order a copy. If you increase the price to p dollars per book you will have fewer orders, and after studying similar book sales you estimate you would have $50000 - 1000p$ sales. What is the largest price that you could set to end up with a revenue of $\$40{,}000$?

108. A new electric car company is ready to sell its debut model. Studying the current state of the electric car market, they estimate that if they charge each customer p dollars, they will have $1000000 - 4p$ sales. What is the lowest price that they could use so that they end up with a revenue of $1 billion?

109. If you add 5 to a certain positive number, then also subtract 2 from that same number, and multiply the results, you get $-\frac{33}{4}$. What was the original number?

110. If you double a certain positive number, then also add $\frac{13}{2}$ to that same number, and multiply the results, you get 24. What was the original number?

Challenge

111. Solve for x in the equation $mx^2 + nx + p = 0$, assuming $m \neq 0$.

Appendices

Appendix A

Unit Conversions

Units of Length in the US/Imperial System	Units of Length in the Metric System	System to System Length Conversions
1 foot (ft) $= 12$ inches (in)	1 meter (m) $= 1000$ millimeters (mm)	1 inch (in) $= 2.54$ centimeters (cm)
1 yard (yd) $= 3$ feet (ft)	1 meter (m) $= 100$ centimeters (cm)	1 meter (m) ≈ 3.281 feet (ft)
1 yard (yd) $= 36$ inches (in)	1 meter (m) $= 10$ decimeters (dm)	1 meter (m) ≈ 1.094 yard (yd)
1 mile (mi) $= 5280$ feet (ft)	1 dekameter (dam) $= 10$ meters (m)	1 mile (mi) ≈ 1.609 kilometer (km)
	1 hectometer (hm) $= 100$ meters (m)	
	1 kilometer (km) $= 1000$ meters (m)	

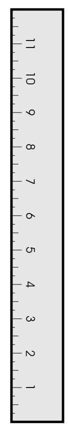


Table A.0.1 Length Unit Conversion Factors

Units of Area in the US/Imperial System	Units of Area in the Metric System	System to System Area Conversions
1 acre $= 43560$ square feet (ft^2)	1 hectare (ha) $= 10000$ square meters (m^2)	1 hectare (ha) ≈ 2.471 acres
640 acres $= 1$ square mile (mi^2)	100 hectares (ha) $= 1$ square kilometer (km^2)	

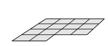


Table A.0.2 Area Unit Conversion Factors

Units of Volume in the US/Imperial System	Units of Volume in the Metric System	System to System Volume Conversions
1 tablespoon (tbsp) $= 3$ teaspoon (tsp)	1 cubic centimeter (cc) $= 1$ cubic centimeter (cm^3)	1 cubic inch (in^3) ≈ 16.39 milliliters (mL)
1 fluid ounce (fl oz) $= 2$ tablespoons (tbsp)	1 milliliter (mL) $= 1$ cubic centimeter (cm^3)	1 fluid ounce (fl oz) ≈ 29.57 milliliters (mL)
1 cup (c) $= 8$ fluid ounces (fl oz)	1 liter (L) $= 1000$ milliliters (mL)	1 liter (L) ≈ 1.057 quarts (qt)
1 pint (pt) $= 2$ cups (c)	1 liter (L) $= 1000$ cubic centimeters (cm^3)	1 gallon (gal) ≈ 3.785 liters (L)
1 quart (qt) $= 2$ pints (pt)		
1 gallon (gal) $= 4$ quarts (qt)		
1 gallon (gal) $= 231$ cubic inches (in^3)		

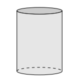


Table A.0.3 Volume Unit Conversion Factors

Units of Mass/Weight in the US/Imperial System	Units of Mass/Weight in the Metric System	System to System Mass/Weight Conversions
1 pound (lb) $= 16$ ounces (oz)	1 gram (g) $= 1000$ milligrams (mg)	1 ounce (oz) ≈ 28.35 grams (g)
1 ton (T) $= 2000$ pounds (lb)	1 kilogram (kg) $= 1000$ grams (g)	1 kilogram (kg) ≈ 2.205 pounds (lb)
	1 metric ton (t) $= 1000$ kilograms (kg)	

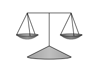


Table A.0.4 Weight/Mass Unit Conversion Factors

Precise Units of Time	Imprecise Units of Time	Units of Time in the Metric System
1 week (wk) $= 7$ days (d)	1 year (yr) ≈ 12 months (mo)	1 second (s) $= 1000$ milliseconds (ms)
1 day (d) $= 24$ hours (h)	1 year (yr) ≈ 52 weeks (wk)	1 second (s) $= 10^6$ microseconds (μs)
1 hour (h) $= 60$ minutes (min)	1 year (yr) ≈ 365 days (d)	1 second (s) $= 10^9$ nanoseconds (ns)
1 minute (min) $= 60$ seconds (s)	1 month (mo) ≈ 30 days (d)	

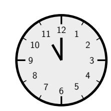


Table A.0.5 Time Unit Conversion Factors

Bytes and Bits	Bytes and Bytes	Bits and Bits
1 byte (B) $= 8$ bits (b)	1 kilobyte (kB) $= 1024$ bytes (B)	1 kilobit (kb) $= 1024$ bits (b)
1 kilobyte (kB) $= 8$ kilobits (kb)	1 megabyte (MB) $= 1024$ kilobytes (kB)	1 megabit (Mb) $= 1024$ kilobits (kb)
1 megabyte (MB) $= 8$ megabits (Mb)	1 gigabyte (GB) $= 1024$ megabytes (MB)	1 gigabit (Gb) $= 1024$ megabits (Mb)
1 gigabyte (GB) $= 8$ gigabits (Gb)	1 terabyte (TB) $= 1024$ gigabytes (GB)	1 terabit (Tb) $= 1024$ gigabits (Gb)

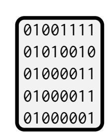


Table A.0.6 Computer Storage/Memory Conversion Factors

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